

Pattern 2: Two Pointers (Same Direction)

The Two Pointers (Same Direction) pattern is one of the most important techniques in Data Structures and Algorithms. It is heavily used in interviews to optimize brute-force solutions involving subarrays, substrings, or continuous elements.

1. What is Two Pointers (Same Direction)?

In this pattern, we use two pointers (usually called Left and Right) that both move forward through the array or string. The pointers maintain a dynamic window that expands and shrinks based on a given condition.

2. Why Do We Use This Pattern?

Brute-force approaches often check all possible subarrays or substrings, resulting in $O(n^2)$ time complexity. The same-direction two pointers technique optimizes this by ensuring each pointer moves at most n times, reducing the time complexity to $O(n)$.

3. When Should This Pattern Click in Your Mind?

- The problem talks about subarray or substring
- Elements are continuous
- You need longest, shortest, or count-based answers
- Conditions like at most, at least, exactly
- In-place removal or compression is required

4. Core Logic / Working

1. Initialize Left = 0 and Right = 0.
2. Move Right forward to expand the window.
3. If the condition is violated, move Left forward to shrink the window.
4. Update the answer whenever the condition is valid.

5. Time & Space Complexity

Time Complexity: $O(n)$, because each pointer moves forward at most n times.

Space Complexity: $O(1)$ or $O(k)$, depending on whether extra data structures like hash maps are used.

6. Common Problems Based on This Pattern

- Longest substring without repeating characters
- Maximum consecutive ones
- Minimum size subarray sum
- Count subarrays with a given condition
- Remove duplicates or elements in-place

7. Common Mistakes to Avoid

- Forgetting to shrink the window after condition violation
- Updating the answer at the wrong time
- Using nested loops unnecessarily
- Confusing this with opposite-direction two pointers

8. One-Line Intuition (Interview Ready)

"Subarray + condition + continuous elements = Two Pointers (Same Direction)."